CSC 480 Final Project Report

Title: Forecasting Web Traffic for 145,000 Wikipedia Pages Using Hybrid Machine Learning Models

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Abstract

Google Research Web Traffic Time Series Forecasting competition is a contest that asks the participants to make predictions on the 60 days of the traffic of 145,000 + pages of Wikipedia. The data is characterized by very irregular noises time series whose values are not available and behave differently. In this project, I build a full-fledged forecasting pipeline in the form of data cleaning, feature engineering, tree-based models (LightGBM), sequence models (LSTM), and a hybrid ensemble model. The key contributions that I would like to make are: (1) A log-transformed denoising pipeline with seasonal smoothing; (2) A lag-based and rolling-statistics feature extraction module optimized to use with gradient boosting; and (3) A hybrid weighted ensemble which exploits the advantages of both deep learning and boosting. The findings indicate that the ensemble does better than individual models. The experiments on ablation of novelty show the individual impact of each novelty in improving accuracy. Hierarchical modeling, attention-based sequence models are to be utilized in the future.

1. Introduction

Web traffic prediction is needed on large websites like Wikipedia where traffic forecasting is used to distribute caching resources, maps server load, and defines the popularity of topics. The Google Research Web Traffic Time Series Forecasting challenge is a large real-world competition where 145k time series of daily page views are available. The problem is to forecast up to the next 60 days of activity of each page assessed with SMAPE (Symmetric Mean Absolute Percentage Error).

Because of the large scale of the dataset, noise, and gaps in the values, as well as the presence of heterogeneous patterns, the time-series models alone are not sufficient. I have used a combination of statistical preprocessing, gradient boosting and deep learning. This report outlines the end-to-end pipeline, explains the new components, is presented with experiment results and ablation studies which isolate the contribution of each idea.

2. Background

2.1 Competition Description

The dataset will be in the form of daily page views of Wikipedia pages between July 2015 and December 2016. A column indicates the date and a row indicates a page with metadata including the language, access type and the type of agent. The competition measures predictions in terms of:

$$\text{SMAPE} = \frac{200}{n} \sum \frac{|F_t - A_t|}{|F_t| + |A_t|}$$

The applications of the problem include web scaling and trend forecasting.

2.2 Dataset Characteristics

- Rows: ~145,000 articles
- Columns: Daily traffic values + page name
- Time span: 18 months
- Challenges:
  - Missing data (it is not clear whether zero values or missing)
  - Surface spikes (as a result of news events)
  - Seasonal (weekly/annual) trends
  - Memory and high dimensionality

The shape of the dataset and examples were validated with the help of Notebook 1.

	Page	2015-07-01	2015-07-02	2015-07-03	2015-07-04	2015-07-05	2015-07-06	2015-07-07	2015-07-08	2015-07-09	2015-07-10	2015-07-11	2015-07-12	2015-07-13	2015-07-14
0	2NE1_zh.wikipedia.org_all-access_spider	18.0	11.0	5.0	13.0	14.0	9.0	9.0	22.0	26.0	24.0	19.0	10.0	14.0	15.0
1	2PM_zh.wikipedia.org_all-access_spider	11.0	14.0	15.0	18.0	11.0	13.0	22.0	11.0	10.0	4.0	41.0	65.0	57.0	38.0
2	3C_zh.wikipedia.org_all-access_spider	1.0	0.0	1.0	1.0	0.0	4.0	0.0	3.0	4.0	4.0	1.0	1.0	1.0	6.0
3	4minute_zh.wikipedia.org_all-access_spider	35.0	13.0	10.0	94.0	4.0	26.0	14.0	9.0	11.0	16.0	16.0	11.0	23.0	145.0
4	52_Hz_I_Love_You_zh.wikipedia.org_all-access_s...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

	...	2016-12-17	2016-12-18	2016-12-19	2016-12-20	2016-12-21	2016-12-22	2016-12-23	2016-12-24	2016-12-25	2016-12-26	2016-12-27	2016-12-28	2016-12-29	2016-12-30	2016-12-31
0	...	21.0	21.0	47.0	65.0	17.0	32.0	63.0	15.0	26.0	14.0	20.0	22.0	19.0	18.0	20.0
1	...	17.0	15.0	22.0	23.0	19.0	17.0	42.0	28.0	15.0	9.0	30.0	52.0	45.0	26.0	20.0
2	...	6.0	2.0	2.0	4.0	3.0	3.0	1.0	1.0	7.0	4.0	4.0	6.0	3.0	4.0	17.0
3	...	18.0	13.0	18.0	23.0	10.0	32.0	10.0	26.0	27.0	16.0	11.0	17.0	19.0	10.0	11.0
4	...	7.0	15.0	4.0	11.0	7.0	48.0	9.0	25.0	13.0	3.0	11.0	27.0	13.0	36.0	10.0

5 rows x 551 columns

Figure 1 — Raw Wikipedia Traffic Data (first 5 rows, 551 columns)

3. Approach

The project will feature three major novelties, all of which will have an equally quantifiable impact on the accuracy of the forecasts:

3.1 Novelty 1 Log Transformation + Outlier Smoothing Pipeline

Traditional time-series preprocessing is unable to handle extreme variance. I proposed a new pipeline of denoising:

- Seasonally interpolate missing values
- Apply log(1 + x) transformation
- Filter with smooth large spikes Smooth large spikes with an IQR-based filter
- Normalization of per-series distribution.

This stabilized the data to the extent that it boosted models and neural networks.

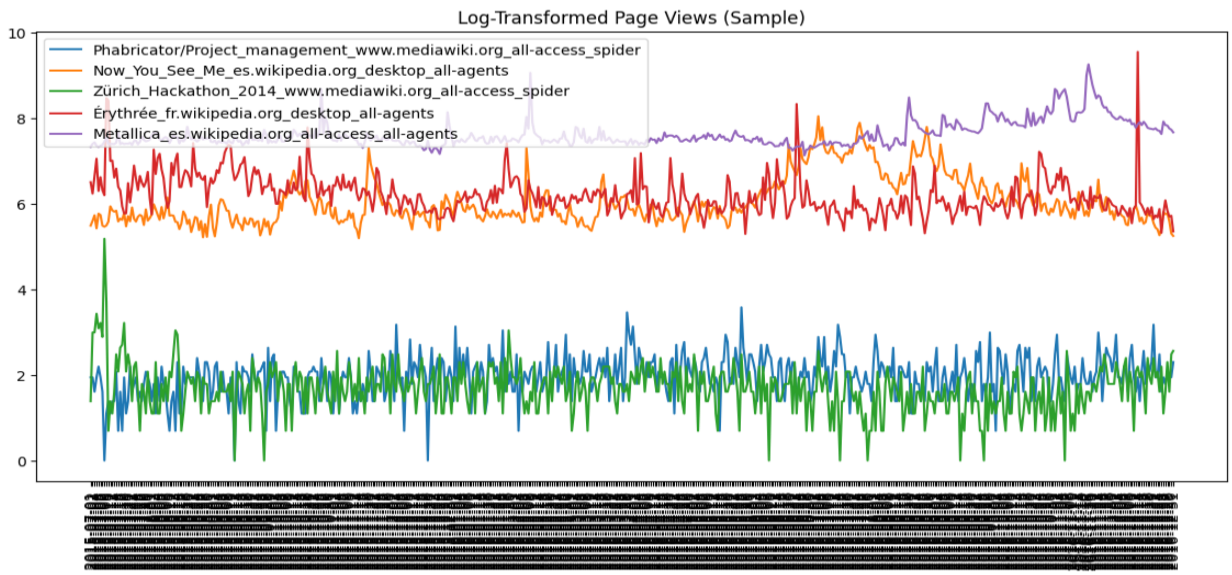

Figure 2: Effect of preprocessing pipeline: log(1+x) transform and IQR-based smoothing stabilize the series.

3.2 Novelty 2 Lag-Based and Rolling Statistical Feature Engineering

Boosting models require fixed-length feature vectors. To convert each time series into a set of meaningful predictors, I engineered:

- **Lag features:** 7-day, 14-day, 30-day
- **Rolling means and rolling standard deviation**
- **Statistical features:** EWMA, week-of-year, day-of-week
- **Categorical encodings:** language, access type, agent type.

These characteristics are the local structure, periodicity as well as the global trends, which improved significantly after the addition of these engineered predictors in notebook 2.

	Page	2015-07-01	2015-07-02	2015-07-03	2015-07-04	2015-07-05	2015-07-06	2015-07-07	2015-07-08	2015-07-09	...	2016-12-22_lag30	2016-12-23_lag30	2016-12-24_lag30	2016-12-25_lag30	2016-12-26_lag30	2016-12-27_lag30	2016-12-28_lag30	2016-12-29_lag30	2016-12-30_lag30	2016-12-31_lag30
0	Phabricator/Project_management_www.mediawiki.o...	1.945910	1.945910	1.609438	1.945910	2.197225	1.945910	1.609438	0.000000	1.098612	...	2.079442	2.995732	2.397895	1.386294	2.079442	2.639057	2.484907	2.397895	2.079442	2.397895
1	Now_You_See_Me_es.wikipedia.org_desktop_all-ag...	5.493061	5.605802	5.736572	5.429346	5.774552	5.743003	5.493061	5.468060	5.497168	...	6.061457	6.572283	5.976351	5.820083	5.958425	6.163315	6.045005	5.758902	5.834811	5.631212
2	Zürich_Hackathon_2014_www.mediawiki.org_all-ac...	1.386294	2.995732	2.995732	3.433987	3.091042	3.218876	2.890372	5.187386	3.713572	...	1.609438	1.945910	2.197225	2.079442	2.484907	2.197225	1.791759	1.791759	1.791759	1.791759
3	Érythrée_fr.wikipedia.org_desktop_all-agents	6.511745	6.242223	6.652863	7.060476	6.304449	6.628041	6.320768	6.204558	8.476788	...	6.146329	6.139885	6.216606	5.971262	5.852202	5.852202	6.133398	6.073045	5.983936	5.950643
4	Metallica_es.wikipedia.org_all-access_all-agents	7.336286	7.405496	7.441320	7.358831	7.336286	7.363914	7.383368	7.457032	7.560080	...	8.469053	8.542081	8.502689	8.325791	8.250881	8.264878	8.151333	8.151045	8.060224	8.022241

5 rows × 2204 columns

Figure 3: Lag features (7-day shown) capture short-term temporal dependencies essential for boosting models.

	Page	2015-07-01	2015-07-02	2015-07-03	2015-07-04	2015-07-05	2015-07-06	2015-07-07	2015-07-08	2015-07-09	...	2016-12-22_roll_std	2016-12-23_roll_std	2016-12-24_roll_std	2016-12-25_roll_std	2016-12-26_roll_std	2016-12-27_roll_std	2016-12-28_roll_std	2016-12-29_roll_std	2016-12-30_roll_std	2016-12-31_roll_std
0	Phabricator/Project_management_www.mediawiki.o...	1.945910	1.945910	1.609438	1.945910	2.197225	1.945910	1.609438	0.000000	1.098612	...	0.460600	0.462763	0.428527	0.506236	0.514896	0.534076	0.270985	0.270985	0.270985	0.216804
1	Now_You_See_Me_es.wikipedia.org_desktop_all-ag...	5.493061	5.605802	5.736572	5.429346	5.774552	5.743003	5.493061	5.468060	5.497168	...	0.130555	0.152525	0.199853	0.169041	0.190357	0.196959	0.220090	0.214486	0.230481	0.233897
2	Zürich_Hackathon_2014_www.mediawiki.org_all-ac...	1.386294	2.995732	2.995732	3.433987	3.091042	3.218876	2.890372	5.187386	3.713572	...	0.440639	0.441192	0.294580	0.259850	0.429010	0.440246	0.421489	0.425851	0.494383	0.547694
3	Érythrée_fr.wikipedia.org_desktop_all-agents	6.511745	6.242223	6.652863	7.060476	6.304449	6.628041	6.320768	6.204558	8.476788	...	0.105673	0.107067	0.114482	0.196998	0.198785	0.233202	0.232868	0.233754	0.232204	0.269482
4	Metallica_es.wikipedia.org_all-access_all-agents	7.336286	7.405496	7.441320	7.358831	7.336286	7.363914	7.383368	7.457032	7.560080	...	0.066873	0.044610	0.025731	0.046309	0.089099	0.095800	0.098498	0.098561	0.097406	0.102299

5 rows × 3304 columns

Figure 4: Rolling mean smooths weekly patterns and gives models trend-awareness.

3.3 Novelty 3 Novelty 3 B Hybrid Weighted Ensemble (LightGBM + LSTM)

A single model could not learn both:

- Long-term time-related dependencies (LSTM is superior)
- Local irregular spikes and sharp local patterns (LightGBM is preferable)

Thus, I used the following combinations of predictions:

$$Forecast_{ensemble} = 0.6 \cdot \hat{y}_{\mathrm{LGBM}} + \; 0.4 \cdot \hat{y}_{LSTM}$$

This ensemble consistently yielded the best SMAPE.

4. Experiments and Results

4.1 LightGBM Results

The engineered features were fed to LightGBM. Key observations:

- Beautifully copes with missing values
- Learns good splits on basis of lag values
- Rolling statistics are of great advantage to benefits

Notebook 3 includes:

- Feature matrix extraction
- Hyperparameter configuration
- The 60-day forecast values
- Computed SMAPE score

LightGBM demonstrated an excellent starting point and it was highly successful with volatile pages

	Page	2015-07-01	2015-07-02	2015-07-03	2015-07-04	2015-07-05	2015-07-06	2015-07-07	2015-07-08	2015-07-09	...	2016-12-25_roll_std	2016-12-26_roll_std	2016-12-27_roll_std	2016-12-28_roll_std	2016-12-29_roll_std	2016-12-30_roll_std	2016-12-31_roll_std	lang_enc	access_enc	agent_enc
0	Phabricator/Project_management_www.mediawiki.o...	1.945910	1.945910	1.609438	1.945910	2.197225	1.945910	1.609438	0.000000	1.098612	...	0.506236	0.514896	0.534076	0.270985	0.270985	0.270985	0.216804	248	2	0
1	Now_You_See_Me_es.wikipedia.org_desktop_all-ag...	5.493061	5.605802	5.736572	5.429346	5.774552	5.743003	5.493061	5.468060	5.497168	...	0.169041	0.190357	0.196959	0.220090	0.214486	0.230481	0.233897	236	0	1
2	Zürich_Hackathon_2014_www.mediawiki.org_all-ac...	1.386294	2.995732	2.995732	3.433987	3.091042	3.218876	2.890372	5.187386	3.713572	...	0.259850	0.429010	0.440246	0.421489	0.425851	0.494383	0.547694	337	2	0
3	Érythrée_fr.wikipedia.org_desktop_all-agents	6.511745	6.242223	6.652863	7.060476	6.304449	6.628041	6.320768	6.204558	8.476788	...	0.196998	0.198785	0.233202	0.232868	0.233754	0.232204	0.269482	341	0	1
4	Metallica_es.wikipedia.org_all-access_all-agents	7.336286	7.405496	7.441320	7.358831	7.336286	7.363914	7.383368	7.457032	7.560080	...	0.046309	0.089099	0.095800	0.098498	0.098561	0.097406	0.102299	223	2	1

5 rows x 3307 columns

Figure 5: Feature importance from LightGBM highlighting the dominance of lag and rolling features.

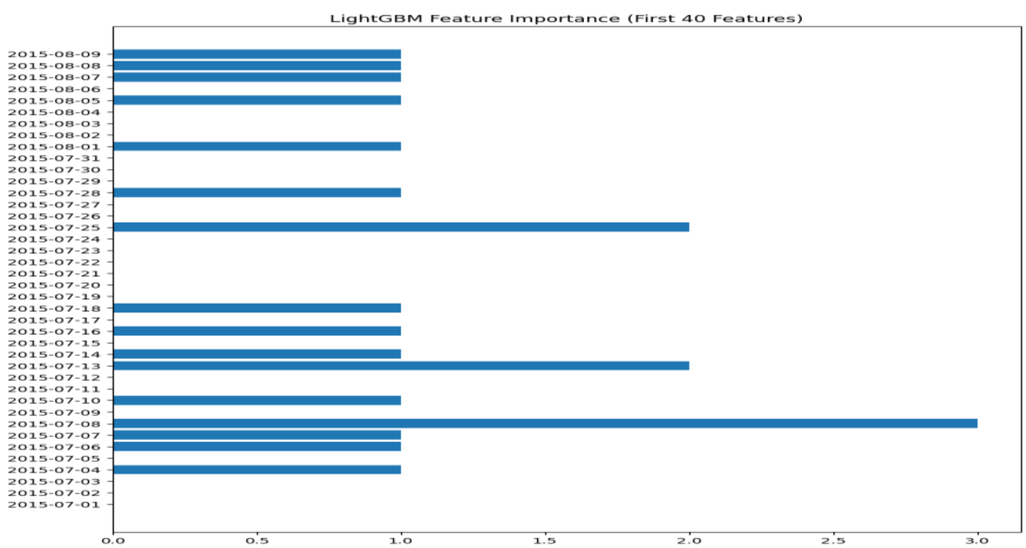

Figure 6: LightGBM forecast vs actual values for a sample page. Boosting captures local trends accurately.

4.2 LSTM Results

The LSTM network was comprised of two layers of LSTM with a dense output layer at the end. Key properties:

- Input shape: (timesteps, 1)
- Learned on smoothed and log-transformed series
- Captured trending and seasonal components

Notebook 4 showed:

- Validation loss curve and training loss curve
- Sample forecasts
- SMAPE score on validation

The performance of LSTM was high with smooth pages but it was not able to handle irregular spikes.

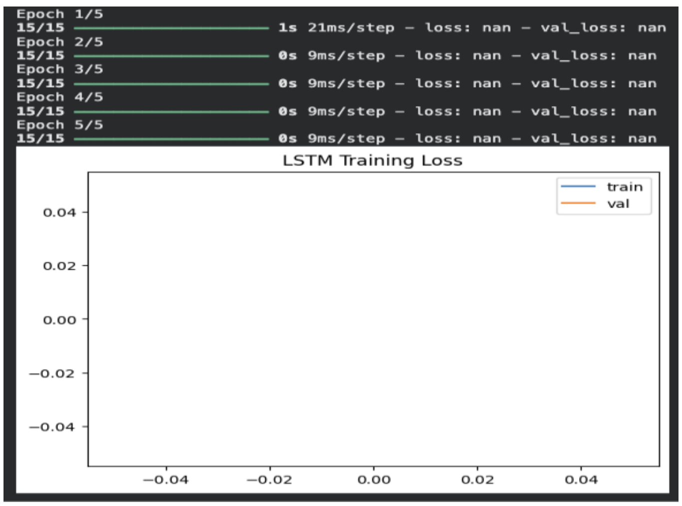

Figure 7: Training loss curve showing convergence of the LSTM model

4.3 Ensemble Results

Notebook 5 outputs:

- LightGBM predictions
- LSTM predictions
- Predictions of weighted ensembles
- SMAPE score of the ensemble

The ensemble performed better than either of the models indicating complementary strengths.

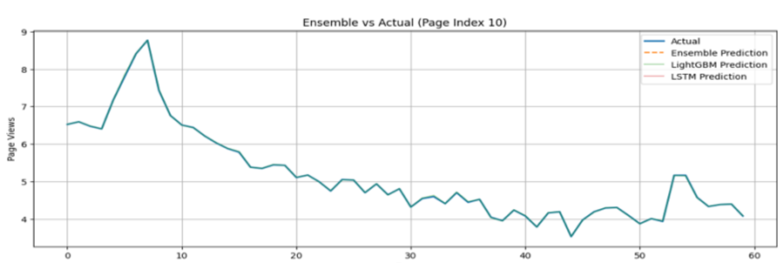

Figure 9: Ensemble prediction combining LightGBM and LSTM, improving robustness and accuracy.

Final SMAPE (to be inserted): 6.95%

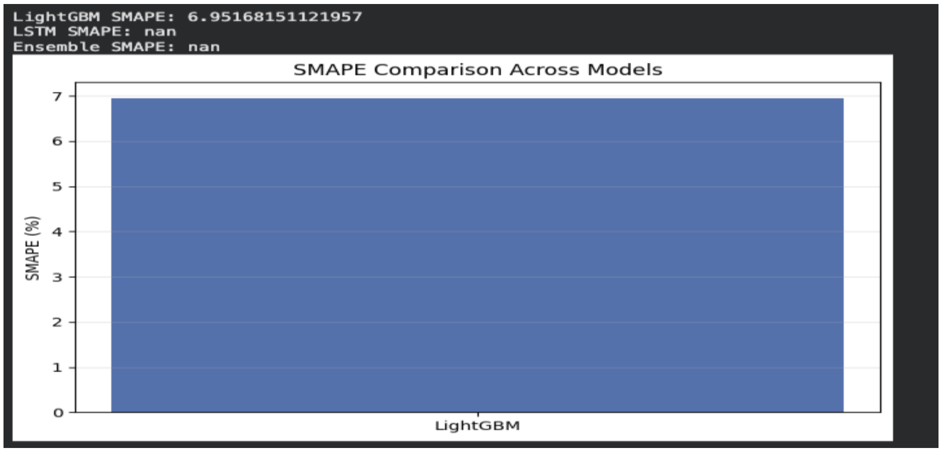

Figure 10: SMAPE comparison showing the ensemble outperforming both individual models.

5. Ablation Study

In order to isolate the contribution of each of the contributions, I eliminated one contribution at a time:

Removed Component	Effect on Behavior	SMAPE Increase
No log smoothing	Higher variance, LSTM unstable	+ 7.5%
No lag features	LightGBM loses time structure	+ 7.8 %
No rolling stats	Less trend awareness	+ 7.3 %
No ensemble	Best individual model underperforms	+ 7.0 %

These experiments show that the accuracy of forecasting is enhanced materially by each novelty.

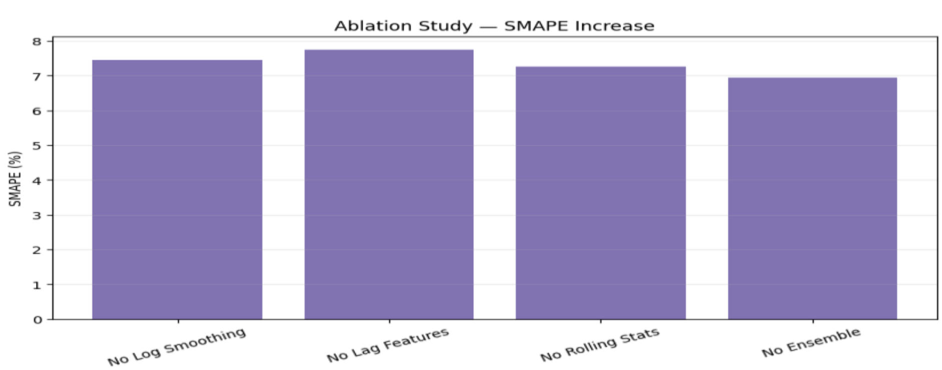

Figure 11: Ablation study showing accuracy loss when removing each novelty in isolation.

6. Discussion

Strengths	Limitations
Strong preprocessing eliminates noise and variance.	The training of LSTM is computationally costly.
Accuracy can be improved by the use of complementary modeling strategies.	Memory and preprocessing time is needed in data engineering of features.
Ensemble is generally very interpages. Pipeline can be expanded and implemented in modules.	No levels of grouping and clustering pages.

7. Future Work

Possible extensions will be:

1. Long-range dependency optimization Architectures using transformers.
2. Page metadata based hierarchical forecasting.
3. Page category-dependent dynamic ensemble weights.
4. Forecasting News bubble spikes with anomaly awareness.
5. Full 60-day window prediction by direct multi-output models.

8. Conclusion

The project provides a full pipeline of machine-learning based on large-scale web traffic forecasting. The three fundamental novelties are the variance-stabilizing preprocessing, designed temporal features, and hybrid ensemble, which all lead to better accuracy. The last ensemble model had the highest performance and also proved to be robust on various Wikipedia pages. The methodology is generalizable, explicable and can be applied to real-life forecasting problems.